Q1. Explain Augmented Reality (AR) and Virtual Reality (VR). (10 Marks)

1. Augmented Reality (AR)

Definition: Augmented Reality (AR) is a technology that adds digital objects, images, sounds, or information to the real-world environment using a smartphone, tablet, or AR glasses.

Main Points:

1. Real-World Environment: AR allows users to see and interact with the real world.
2. Digital Objects: It adds virtual objects, text, images, or animations to the real environment.
3. Real-Time Interaction: Digital content appears and responds in real time.
4. Devices Used: Smartphones, tablets, cameras, and AR glasses are commonly used.
5. Object Recognition: AR can use cameras and sensors to detect surfaces or objects.
6. Applications: It is used in education, gaming, healthcare, shopping, and navigation.

Example: Pokémon GO displays virtual Pokémon characters on the real-world view through a smartphone camera.

2. Virtual Reality (VR)

Definition: Virtual Reality (VR) is a technology that creates a computer-generated, three-dimensional environment in which users feel as if they are present inside a virtual world.

Main Points:

1. Virtual Environment: VR creates a completely computer-generated environment.
2. Immersion: It makes users feel as if they are inside another world.
3. 3D Experience: Users can explore virtual objects and environments in three dimensions.
4. Devices Used: VR headsets, motion controllers, and motion-tracking sensors are commonly used.
5. Interaction: Users can move their heads, hands, or bodies to interact with the virtual environment.
6. Applications: It is used in gaming, training, education, medical simulations, and virtual tours.

Example: A person wearing a VR headset can experience a virtual roller-coaster ride without visiting an actual amusement park.

3. Difference Between AR and VR

Augmented Reality (AR)	**Virtual Reality (VR)**
1. Adds digital content to the real world.	Creates a completely virtual environment.
2. The real world remains visible.	The real world is mostly blocked from view.
3. Combines real and virtual objects.	Provides a computer-generated experience.
4. Uses smartphones, tablets, and AR glasses.	Mainly uses VR headsets and controllers.
5. Provides partial immersion.	Provides a highly immersive experience.
6. Example: Pokémon GO.	Example: VR roller-coaster simulation.

4. Applications of AR and VR

- Education: Interactive learning and virtual classrooms.

- Healthcare: Medical training and surgical simulations.
- Gaming: Interactive AR games and immersive VR games.
- Shopping: Virtual product previews and furniture placement.
- Training: Flight simulators and industrial safety training.

Q2. Explain the Different Types of AR and VR. (10 Marks)

1. Types of Augmented Reality (AR)

Definition: Augmented Reality (AR) adds digital objects, images, or information to the real-world environment. AR is classified into the following types.

1. Marker-Based AR

- It uses a specific marker, such as a QR code or a printed image, to display digital content.
- The camera detects the marker and displays a virtual object.
- Example: Scanning a picture in a textbook to display a 3D model.

2. Markerless AR

- It does not require a predefined marker or QR code.
- It uses sensors, GPS, and computer vision to place virtual objects.
- Example: Placing virtual furniture in a real room using a smartphone.

3. Location-Based AR

- It uses GPS, a compass, and other sensors to provide location-based digital information.
- Virtual content appears according to the user's geographical location.
- Example: Pokémon GO displaying virtual Pokémon at different locations.

4. Projection-Based AR

- It projects digital images or light onto real-world surfaces.
- Users can interact with projected information in some systems.
- Example: Projecting a virtual keyboard onto a table.

2. Types of Virtual Reality (VR)

Definition: Virtual Reality (VR) creates a computer-generated environment that allows users to experience and interact with a virtual world.

1. Non-Immersive VR

- Users interact with a virtual environment through a computer screen.
- They can control the environment using a keyboard, mouse, or controller.
- Example: Playing a 3D computer game on a monitor.

2. Semi-Immersive VR

- It provides a partial feeling of being inside a virtual environment.
- It often uses large screens, projection systems, or simulators.
- Example: A flight simulator used for pilot training.

3. Fully Immersive VR

- It provides a highly realistic and immersive virtual experience.
- It uses VR headsets, motion tracking, and controllers.
- Users can look around and interact with virtual objects.
- Example: Playing a VR game using a Meta Quest headset.

3. Difference Between AR and VR Types

Augmented Reality (AR)	Virtual Reality (VR)
Adds digital content to the real world.	Creates a computer-generated world.
Real surroundings remain visible.	The virtual environment becomes the main experience.
Uses cameras, GPS, and sensors.	Uses headsets, controllers, and tracking systems.
Example: Marker-based AR.	Example: Fully immersive VR.

4. Applications

- Education: Interactive learning and virtual classrooms.
- Gaming: AR mobile games and immersive VR games.
- Healthcare: Medical training and simulations.
- Industry: Product visualization and employee training.
- Navigation: Location-based AR directions.

Q3. Explain the Applications and Use Cases of AR and VR. (10 Marks)

1. Introduction

Augmented Reality (AR) adds digital information, images, and 3D objects to the real-world environment.

Virtual Reality (VR) creates a computer-generated environment in which users can interact with virtual objects and experience an artificial world.

Both technologies are used in education, healthcare, gaming, business, and many other fields.

2. Applications of Augmented Reality (AR)

1. Education and Training

- AR displays 3D models of the human body, planets, and machines.
- It makes learning more interactive and easier to understand.
- Example: Viewing a 3D model of the solar system through a smartphone.

2. Healthcare

- AR helps doctors visualize medical information and anatomical structures.
- It can support medical training and certain surgical procedures.
- Example: Displaying 3D anatomical information during medical training.

3. Gaming and Entertainment

- AR combines virtual characters with real-world surroundings.

- It provides an interactive gaming experience.
- Example: Pokémon GO.

4. Shopping and Retail

- Customers can preview products before purchasing them.
- AR helps users check how furniture, glasses, or other products look in real life.
- Example: Placing a virtual sofa in a room using a mobile application.

5. Navigation and Tourism

- AR displays directions, landmarks, and useful information over a camera view.
- It helps users find places and explore tourist destinations.
- Example: Displaying directional arrows on a smartphone screen.

3. Applications of Virtual Reality (VR)

1. Gaming and Entertainment

- VR creates an immersive 3D environment.
- Users can interact with virtual objects using controllers.
- Example: Playing a VR racing game with a headset.

2. Education and Training

- VR allows students to experience virtual classrooms and practical simulations.
- It helps learners practise without real-world risks.
- Example: Training pilots using a flight simulator.

3. Healthcare

- VR is used in medical education and simulated surgical practice.
- It can also support certain rehabilitation and psychological treatments under professional guidance.
- Example: Medical students practising a simulated operation.

4. Business and Architecture

- VR allows users to explore buildings and product designs before construction or production.
- It helps identify design problems and improve planning.
- Example: Taking a virtual tour of a house that has not yet been built.

5. Military and Industrial Training

- VR provides simulated environments for practising difficult or dangerous tasks.
- It reduces the need to perform every training activity in real conditions.
- Example: Soldiers practising emergency situations in a virtual battlefield.

4. Difference Between AR and VR Applications

AR applications	VR applications
Virtual try-on for glasses and makeup	Immersive games
Camera-based educational models	Virtual classrooms
Navigation and tourism information	Flight simulators
Snapchat face filters	Virtual building tours
Virtual furniture placement	Simulated training environments

5. Benefits of AR and VR

1. They make learning more interactive.
2. They help explain difficult concepts visually.
3. They improve user engagement.
4. They allow users to preview products and environments.
5. They can reduce the need for some physical training setups.

6. Limitations of AR and VR

1. Some devices are expensive.
2. Advanced features may require powerful hardware.
3. Certain VR experiences can cause dizziness or discomfort.
4. Creating high-quality AR/VR content can require technical skills.

Q4. Explain Lens Studio and the Steps to Publish a Lens. (10 Marks)

1. Definition of Lens Studio

Lens Studio is a free Augmented Reality (AR) development tool provided by Snapchat. It allows users to create interactive AR effects, filters, animations, and 3D experiences for Snapchat.

2. Features of Lens Studio

1. AR Filters: Create face filters, masks, and special camera effects.
2. 3D Objects: Add 3D models, animations, and virtual objects to a scene.
3. Face Tracking: Detect and track a user's face to apply effects.
4. World Tracking: Place virtual objects on real-world surfaces.
5. Scripting: Use JavaScript to create interactive experiences.
6. Preview and Testing: Test the Lens on a computer and compatible mobile devices before publishing.

3. Steps to Publish a Lens

Step 1: Download and Install

Download Lens Studio from the official website and sign in using your Snapchat account.

Step 2: Create a Project

Open Lens Studio and create a new project using a template or start from scratch.

Step 3: Design the Lens

Add images, 3D objects, face masks, animations, sounds, or other AR effects.

Step 4: Preview and Test

Preview the Lens, test its features, and fix errors to ensure it works correctly.

Step 5: Publish the Lens

Click Publish Lens, provide the required details such as the name and icon, and submit the Lens for review.

Step 6: Review and Share

After approval, share the Lens using its Snapcode or link. Other users can open and try it on Snapchat.

4. Applications of Lens Studio

- Social Media: Creating Snapchat filters and effects.
- Entertainment: Developing interactive AR games.
- Marketing: Promoting brands and products using AR filters.
- Education: Creating interactive 3D learning experiences.
- Creativity: Designing customized face effects and animations.

Install → Create Project → Design → Test → Publish → Review and Share

Q5. Explain Lens Size, Text Size and App Icon in Lens Studio. (10 Marks)

1. Lens Size

Definition: Lens Size refers to the amount of storage space occupied by an AR Lens. It is important because a smaller Lens generally loads faster and provides a better user experience.

Main Points:

1. File Size: It represents the total size of the Lens assets and resources.
2. Loading Speed: A smaller Lens usually downloads and loads faster.
3. Performance: Optimized assets can help the Lens run smoothly on mobile devices.
4. Asset Optimization: Reduce the size of 3D models, textures, audio files, and animations.
5. Compression: Use suitable compression for images and other supported assets.
6. Testing: Check the Lens size and performance before publishing.

Example: Compressing a large texture and reducing unnecessary 3D models can reduce the Lens size.

2. Text Size

Definition: Text Size refers to the size of the text displayed within an AR Lens. Proper text size makes the Lens easy to read on mobile screens.

Main Points:

1. Readability: Text should be clear and easy to read.
2. Font Size: Choose an appropriate font size for the target screen.
3. Visibility: Text should contrast with the background.
4. Placement: Position text so it does not cover important objects or the user's face.
5. Consistency: Use consistent font styles and sizes throughout the Lens.

6. Testing: Preview text on different screen sizes to check its readability.

Example: In a birthday Lens, the text “Happy Birthday” should be large enough to read clearly on a smartphone.

3. App Icon

Definition: An App Icon, in the Lens publishing context, is the small visual image or thumbnail used to represent a Lens and help users identify it.

Main Points:

1. Identification: It helps users recognize the Lens.
2. Visual Design: The icon should be simple, attractive, and relevant to the Lens.
3. Image Quality: Use a clear image that remains recognizable at a small size.
4. Relevance: The icon should represent the effect or purpose of the Lens.
5. Publishing: Add the required icon or thumbnail in the publishing settings, following the current platform requirements.
6. Branding: A suitable icon can help create a consistent visual identity.

Example: A Lens that adds puppy ears can use a clear thumbnail showing the puppy-ear effect.

4. Difference Between Lens Size, Text Size and App Icon

Feature	Meaning	Main Purpose
Lens Size	Storage space used by the Lens	Faster loading and optimization
Text Size	Size of displayed text	Better readability
App Icon	Image representing the Lens	Identification and recognition

5. Advantages

- A suitable Lens size helps improve loading and performance.
- Proper text size makes information easy to read.

Q6. Explain How to Add Text in Lens Studio. (10 Marks)

1. Definition

Lens Studio is an AR development tool by Snapchat that allows users to create interactive effects. Text can be added to a Lens to display messages, names, instructions, labels, or other information in an Augmented Reality environment.

2. Steps to Add Text in Lens Studio

Step 1: Open Lens Studio

Open Lens Studio and create a new project or open an existing project.

Step 2: Add a Text Object

Open the Objects or Scene Hierarchy panel, click the Add Object (+) option, and select the available text object type.

Step 3: Enter the Text

Select the text object and use its Inspector or text properties to enter the required message, such as “Hello World”.

Step 4: Customize the Text

Adjust the font, size, color, alignment, and other available properties to make the text attractive and readable.

Step 5: Set Position and Size

Use the transform controls to move, rotate, and scale the text so that it appears in the correct position.

Step 6: Preview and Test

Preview the Lens to check the text's appearance, visibility, position, and readability. Make changes if required.

3. Important Properties of Text

1. Text Content: Defines the words or message displayed.
2. Font Style: Controls the appearance of the letters.
3. Font Size: Determines how large the text appears.
4. Text Color: Changes the color of the text.
5. Alignment: Positions text to the left, center, or right where supported.
6. Transform: Controls the text object's position, rotation, and scale.

4. Example

Suppose we want to create a birthday AR Lens.

- Add a text object to the project.
- Enter the text “Happy Birthday!”
- Set the text color to pink and choose a suitable font size.
- Place the text above the person's head.
- Preview the Lens to check the final result.

Open Project → Add Text → Enter Content → Customize → Position → Preview

Q7. Explain How to Add Audio and Video in Lens Studio. (10 Marks)

1. Introduction

Lens Studio is an Augmented Reality (AR) development tool by Snap Inc. It allows developers to create interactive Lenses by adding visual effects, 3D objects, audio, and supported video content.

2. How to Add Audio in Lens Studio

Definition: Audio is used to add background music, sound effects, or voice recordings to an AR Lens to make it more interactive.

Step 1: Open the Project

Open Lens Studio and create a new project or open an existing one.

Step 2: Import Audio

Import a supported audio file, such as WAV, using the asset or Resources panel.

Step 3: Add an Audio Component

Add the appropriate audio component or audio controller to the scene and assign the imported audio asset.

Step 4: Configure and Test

Adjust available settings such as volume and playback behavior. Preview the Lens to check that the audio works correctly.

3. How to Add Video in Lens Studio

Definition: Video adds moving visual content to an AR Lens. Depending on the Lens Studio version and supported workflow, video can be used as a texture or displayed through an appropriate video component.

Step 1: Open the Project

Open Lens Studio and create a new project or load an existing project.

Step 2: Import the Video

Import a supported video asset in a format accepted by the current Lens Studio workflow.

Step 3: Create a Display Surface

Add a suitable object, such as a plane, and configure the supported video texture or playback component to display the video.

Step 4: Configure and Test

Set the available playback options, check the video's appearance, and preview the Lens to confirm that it works properly.

4. Advantages of Adding Audio and Video

1. Better User Experience: Makes the Lens more attractive and engaging.
2. Realistic Effects: Audio and video make AR scenes feel more natural.
3. Entertainment: Supports creative filters and interactive experiences.
4. Communication: Voice and visual content can explain information clearly.
5. Marketing: Helps create engaging promotional Lenses.

5. Example

Suppose we create a birthday Lens:

- Add a birthday background or visual effect.
- Import a birthday song and configure its playback.
- Add a supported video element showing animated balloons.
- Preview the Lens and check the audio and video.
- Publish the Lens after testing and completing the required review.

Remember these sequences:

→Audio: Open Project → Import Audio → Add Audio Component → Configure → Test

→Video: Open Project → Import Video → Add Display Surface → Configure → Test

Q8. Explain World Mesh in Lens Studio. (10 Marks)

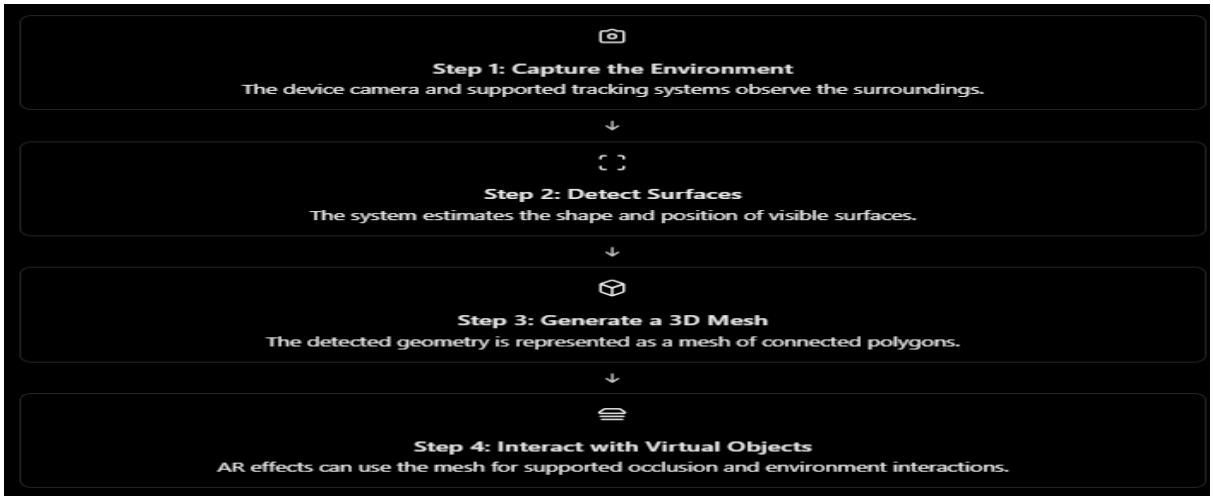
1. Definition

World Mesh is an Augmented Reality (AR) feature that creates a 3D mesh representation of the real-world environment. It helps an AR application understand physical surfaces and surroundings so that virtual objects can interact more realistically with the real world.

2. Features of World Mesh

1. 3D Environment Mapping: World Mesh represents detected real-world surfaces using a collection of connected polygons.
2. Surface Detection: It helps represent surfaces such as floors, walls, tables, and other objects.
3. Realistic Occlusion: It can help virtual objects appear behind real-world surfaces when the appropriate occlusion setup is used.
4. Physics Interaction: Virtual objects can interact with the detected environment when suitable physics components are configured.
5. Real-Time Updates: The mesh may update as the device scans or observes its surroundings.
6. Improved AR Experience: It helps virtual objects fit more naturally into the physical environment.

3. How World Mesh Works in Lens Studio



4. Applications of World Mesh

1. Gaming: Virtual characters and objects can interact with real-world surroundings.
2. Occlusion: Virtual objects can appear hidden behind detected physical surfaces.
3. Interior Design: AR objects can be positioned more naturally in rooms.
4. Education: Students can explore 3D models in a real environment.
5. AR Experiences: Developers can create more realistic interactions between virtual objects and physical surroundings.

5. Example

Suppose we create an AR game in Lens Studio in which a virtual ball falls onto the floor.

- The device scans the surrounding environment.
- World Mesh represents the detected floor surface.
- A suitable physics setup allows the virtual ball to interact with that surface.

- The ball appears to land on the real floor instead of passing through it.

6. Advantages of World Mesh

- Improves environmental understanding.
- Supports more realistic AR effects.
- Helps create natural interactions between real and virtual objects.
- Can improve occlusion and surface-based effects.
- Makes AR experiences more immersive.

Q10. Explain Hand Tracking in Lens Studio. (10 Marks)

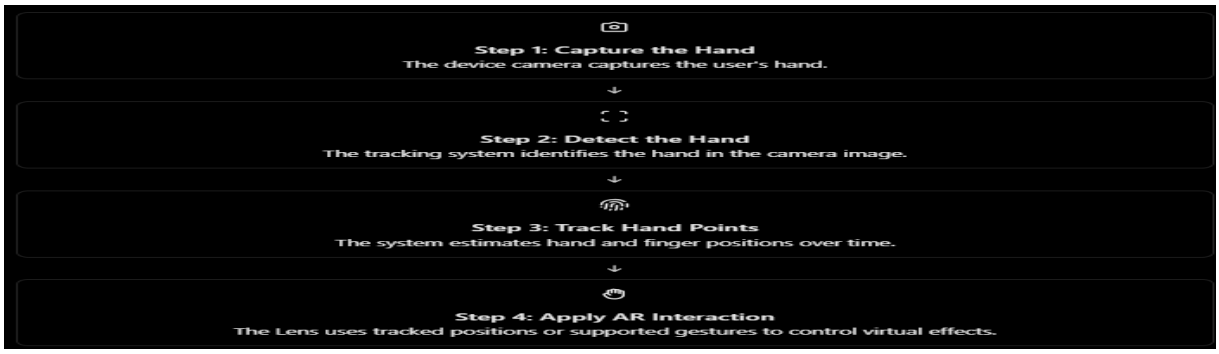
1. Definition

Hand Tracking is an Augmented Reality (AR) technology that detects and tracks a user's hands using a camera and computer vision. In Lens Studio, it can be used to create interactive effects that respond to hand movements and gestures.

2. Features of Hand Tracking

1. Hand Detection: Detects the user's hand in the camera view.
2. Finger Tracking: Identifies fingers and estimates their positions.
3. Gesture Recognition: Supports interactions based on hand gestures, depending on the tracking system.
4. Real-Time Tracking: Updates hand positions as the user moves.
5. AR Effects: Allows virtual objects and effects to follow or respond to hands.
6. Interactive Experience: Enables users to control AR effects without touching the screen in supported experiences.

3. How Hand Tracking Works in Lens Studio



4. Steps to Use Hand Tracking in Lens Studio

1. Open Lens Studio and create a new project.
2. Add a supported hand-tracking object or template from the Objects panel.
3. Configure the available hand-tracking settings.
4. Connect tracked hand positions or gestures to the desired AR objects or effects.
5. Preview the Lens and test different hand movements.
6. Optimize and publish the Lens after testing.

5. Applications of Hand Tracking

1. Gaming: Control virtual characters using hand movements.
2. Education: Interact with virtual 3D models using gestures.
3. Entertainment: Apply AR effects to hands and fingers.
4. Virtual Controls: Move, rotate, or select virtual objects without physical buttons.
5. Training: Demonstrate hand movements and gestures in interactive simulations.

6. Example

Suppose we create an AR Lens that displays a glowing effect around a user's hand.

- The camera captures the hand.
- Hand tracking identifies the hand and estimates its position.
- The Lens attaches a glowing visual effect to the tracked hand.
- When the user moves their hand, the effect follows it.

7. Advantages of Hand Tracking

- Provides natural, touch-free interaction.
- Makes AR applications more engaging.
- Supports gesture-based controls.
- Improves interactive gaming and learning.
- Reduces the need for physical controllers in supported experiences.

Q11. Explain Hand Joints in Lens Studio. (10 Marks)

1. Definition

Hand Joints are the key points on a hand that represent the positions of the wrist, fingers, and finger joints. In Lens Studio, hand-tracking systems can use these points to track hand movements and create interactive Augmented Reality (AR) effects.

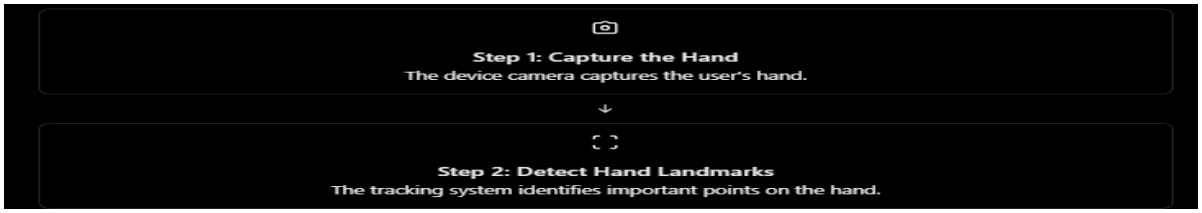
2. Types of Hand Joints

Hand joints represent different parts of the hand:

1. Wrist: The point connecting the hand to the arm.
2. Thumb Joints: Points representing the thumb's base and bending positions.
3. Index Finger Joints: Points representing the base, middle, and tip of the index finger.
4. Middle Finger Joints: Points representing the bending positions and tip of the middle finger.
5. Ring Finger Joints: Points representing the joints and tip of the ring finger.
6. Little Finger Joints: Points representing the joints and tip of the little finger.

The exact number and naming of joints depend on the hand-tracking system used.

3. How Hand Joints Work in Lens Studio





4. Applications of Hand Joints

1. AR Jewellery: Place virtual rings on fingers.
2. Hand Effects: Add glowing lines or animations to fingers.
3. Gesture Recognition: Identify certain hand poses using joint positions.
4. Virtual Controls: Use finger movements to control virtual objects.
5. Education: Demonstrate hand anatomy and finger movements.
6. Gaming: Control game elements using hand movements.

5. Example

Suppose we create an AR Lens that displays a virtual ring on the index finger.

- The camera detects the user's hand.
- The tracking system identifies the index finger joints.
- The Lens uses the appropriate finger position to place the virtual ring.
- When the finger moves, the ring follows it.

6. Advantages of Hand Joints

- Provide detailed information about hand position.
- Help create accurate hand-attached AR effects.
- Support gesture-based interaction.
- Make AR applications more realistic.
- Improve tracking of finger movements.

Q12. Explain the Difference Between 2D and 3D Hand Tracking. (10 Marks)

1. Definition

2D Hand Tracking: 2D hand tracking detects and tracks the position of a hand and its fingers in a two-dimensional image using X and Y coordinates.

3D Hand Tracking: 3D hand tracking detects and tracks the hand in three-dimensional space using X, Y, and Z coordinates, including depth information.

2. Difference Between 2D and 3D Hand Tracking

Basis	2D Hand Tracking	3D Hand Tracking
1. Coordinates	Uses X and Y coordinates.	Uses X, Y, and Z coordinates.
2. Dimension	Works in two dimensions.	Works in three dimensions.

3. Depth	Does not directly measure depth.	Estimates or measures depth.
4. Data	Tracks hand positions on a flat image.	Tracks hand positions in 3D space.
5. Accuracy	Limited for depth-related movements.	Better for understanding depth and spatial movement.
6. Technology	Usually uses a normal camera and computer vision.	May use depth cameras, multiple cameras, or AI-based 3D estimation.
7. Complexity	Generally simpler and faster to process.	Generally more complex and computationally demanding.
8. Applications	Gesture recognition, image filters, and simple hand effects.	VR interaction, 3D object manipulation, and realistic AR effects.

3. Working of 2D Hand Tracking

1. The camera captures an image of the hand.
2. The system detects the hand and its key points.
3. It identifies positions using X and Y coordinates.
4. The system tracks finger movements and recognizes gestures.
5. An AR effect or action is applied based on the hand movement.

Example: A filter follows the user's hand on the screen.

4. Working of 3D Hand Tracking

1. The camera or tracking system captures hand movements.
2. The system detects the hand and its joints.
3. It estimates the X, Y, and Z coordinates of the hand joints.
4. It tracks hand movement, depth, and orientation.
5. A 3D object or virtual effect responds to the hand's movement in space.

Example: A user moves or rotates a virtual 3D object using hand gestures in VR.

5. Applications

2D Hand Tracking

- Snapchat and AR hand filters.
- Basic gesture recognition.
- Interactive camera applications.
- Simple touch-free controls.

3D Hand Tracking

- Virtual Reality (VR) games.
- Augmented Reality (AR) object manipulation.
- Virtual training and simulations.
- 3D modeling and gesture-based controls.

Q13. Explain Body Tracking in Lens Studio. (10 Marks)

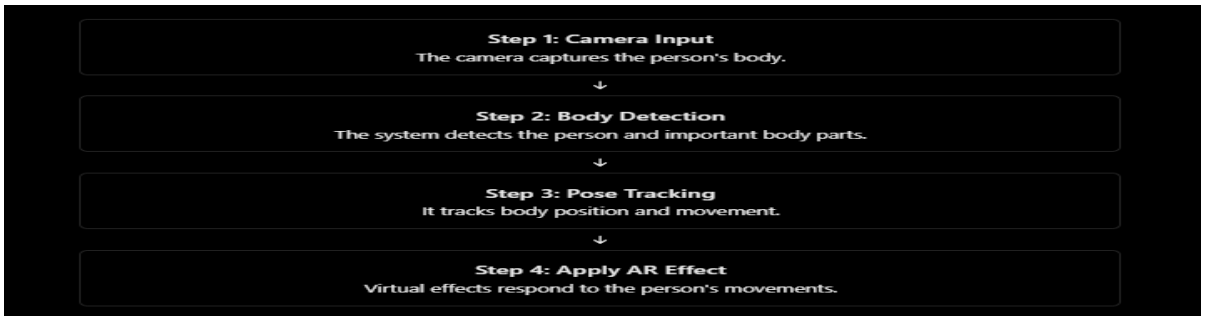
1. Definition

Body Tracking is a technology that detects and tracks a person's body movements using a camera and computer vision. In Lens Studio, it helps AR effects respond to body position, posture, and movement.

2. Features of Body Tracking

1. Body Detection: Detects a person's body in the camera view.
2. Pose Estimation: Identifies important body parts such as shoulders, elbows, knees, and hips.
3. Movement Tracking: Tracks body movements in real time.
4. AR Effects: Allows virtual objects and effects to follow the body.
5. Gesture Recognition: Can support interactions based on body poses and movements.
6. Real-Time Response: Updates the effect as the person moves.

3. Working of Body Tracking



4. Steps to Use Body Tracking in Lens Studio

1. Open Lens Studio and create a new project.
2. Import the required 3D models, textures, or visual effects.
3. Add the supported body-tracking feature or template available in your Lens Studio version.
4. Configure the effect to respond to the tracked body or pose.
5. Adjust the position, size, and appearance of the effect.
6. Preview and test the Lens using the available preview or device-testing options.

Note: Available body-tracking features and setup steps may vary by Lens Studio version.

5. Applications of Body Tracking

- AR Filters: Adds visual effects to a person's body.
- Gaming: Controls game actions through body movements.
- Fashion: Places virtual clothing or accessories on a person.
- Education: Demonstrates body posture and movement.
- Entertainment: Creates interactive videos and social media effects.

6. Example

Suppose a user creates an AR Lens that displays glowing effects around a person's body. When the person moves their arms or changes their pose, the effect responds to the tracked body movement.

7. Advantages

1. Provides interactive AR experiences.
2. Makes effects respond naturally to body movements.
3. Supports hands-free interaction.
4. Improves gaming and entertainment experiences.

Q14. Explain Body Mesh in Lens Studio. (10 Marks)

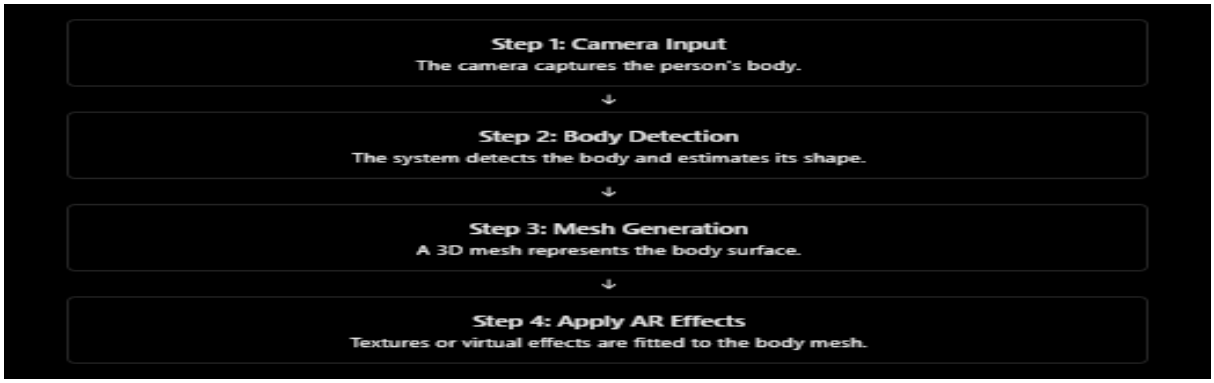
1. Definition

Body Mesh is a 3D representation of a person's body made up of connected points and surfaces. In AR applications, it helps place virtual effects and materials on the body according to its shape and movement.

2. Features of Body Mesh

1. 3D Body Representation: Creates a virtual representation of the person's body.
2. Mesh Structure: Uses vertices, edges, and faces to represent body surfaces.
3. Body Fitting: Allows visual effects to fit the person's body.
4. Movement Support: The mesh can update as the person moves, depending on the tracking system.
5. AR Effects: Supports effects such as virtual clothing, body colors, and textures.
6. Real-Time Interaction: Helps AR effects respond to body movement.

3. Working of Body Mesh



4. Steps to Use Body Mesh in Lens Studio

1. Open Lens Studio and create a new project.
2. Select a suitable body-related template or supported tracking feature.
3. Add the required body mesh or body-effect resources.
4. Apply a material, texture, or visual effect to the mesh.
5. Adjust the material, scale, and appearance as required.
6. Preview and test the Lens to check how the effect follows the body.

Note: The exact components and workflow depend on the Lens Studio version and the supported tracking features.

5. Applications of Body Mesh

- Virtual Clothing: Places virtual clothes or patterns on a body.
- AR Filters: Applies colors, textures, or visual effects to the body.

- Gaming: Creates interactive body-based effects.
- Fashion and Shopping: Demonstrates virtual outfit designs.
- Entertainment: Creates creative body effects for videos.
- Education: Helps visualize body shape and structure.

6. Example

Suppose a user creates an AR Lens that applies a glowing texture to a person's T-shirt area. The body mesh helps the texture fit the body surface and move with the person.

7. Difference Between Body Tracking and Body Mesh

Body Tracking	Body Mesh
Detects and tracks body position and movement.	Represents the body's surface as a 3D mesh.
Focuses on movement and pose.	Focuses on body shape and surface effects.
Can use body landmarks or joints.	Uses mesh geometry such as vertices and faces.

8. Advantages

1. Helps create realistic body-based AR effects.
2. Allows textures to fit body surfaces.
3. Improves interactive AR experiences.
4. Supports creative fashion and entertainment applications.

Q15. Explain Behaviour Script in Lens Studio. (10 Marks)

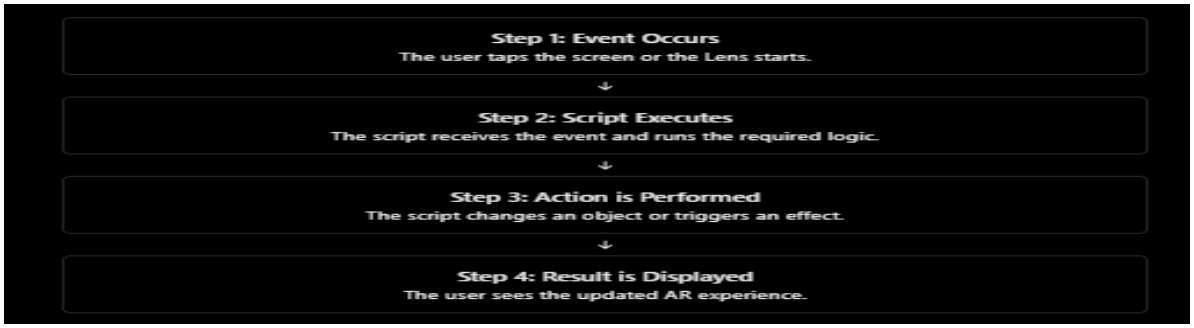
1. Definition

A Behaviour Script in Lens Studio is a script used to control how objects and AR effects behave when specific events or conditions occur. It helps developers add interactivity to a Lens using programming.

2. Features of Behaviour Script

1. Event Handling: Performs actions when an event occurs, such as a screen tap.
2. Object Control: Changes an object's position, rotation, or scale.
3. Animation Control: Starts or stops animations.
4. User Interaction: Makes AR effects respond to user actions.
5. Conditional Logic: Uses if, else, and other programming statements to control actions.
6. Automation: Performs tasks automatically when the Lens starts or an event occurs.

3. Working of Behaviour Script



4. Steps to Add a Script in Lens Studio

1. Open Lens Studio and create a project.
2. Select the object that requires interactive behaviour.
3. Add a Script component or create a script asset.
4. Write the required JavaScript code.
5. Connect the script to the required scene objects or events.
6. Save the project and preview the Lens.
7. Test the interaction and correct any errors.

5. Example of Behaviour Script

The following JavaScript example toggles an object's visibility when the user taps the screen. It illustrates the logic; the input event and object reference must be connected appropriately in Lens Studio.

```
// Reference to the object
var obj = script.myObject;
// Toggle object visibility
function toggleObject() {
  obj.enabled = !obj.enabled;
}
```

Explanation:

- obj refers to the object controlled by the script.
- enabled determines whether the object is active.
- ! reverses the current true/false value.
- When toggleObject() is called, the object's visibility state changes.

6. Applications of Behaviour Script

- Interactive Filters: Changes effects when users tap the screen.
- Animation: Starts or stops animations.
- Gaming: Controls virtual objects and game actions.
- Audio Effects: Plays sounds in response to events.
- Object Manipulation: Changes the position, scale, or rotation of objects.
- User Interface: Controls buttons and interactive elements.

7. Advantages

1. Adds interactivity to AR experiences.
2. Automates actions using scripts.
3. Allows flexible control of scene objects.
4. Improves user engagement.
5. Supports customized AR effects.

Q16. Explain Tween Script in Lens Studio. (10 Marks)

1. Definition

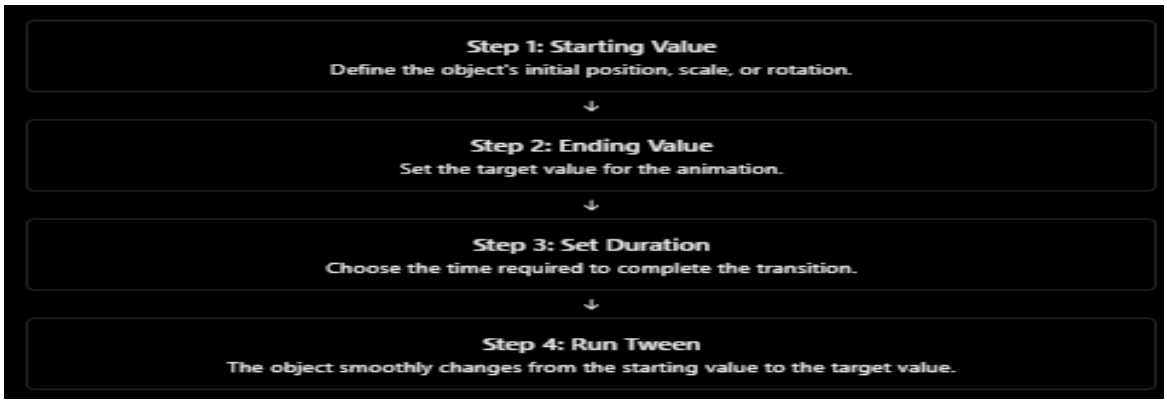
A Tween Script in Lens Studio is used to create smooth transitions between two values over a specific period of time. It can animate an object's position, rotation, scale, color, or other properties.

For example, a 3D object can smoothly move from one position to another instead of moving suddenly.

2. Features of Tween Script

1. Smooth Animation: Creates smooth movement between starting and ending values.
2. Position Animation: Moves an object from one position to another.
3. Scale Animation: Increases or decreases the size of an object.
4. Rotation Animation: Rotates an object smoothly.
5. Duration Control: Defines how long the animation takes.
6. Easing Functions: Controls the animation style, such as slow start or slow finish.
7. Interactive Effects: Can animate objects in response to user actions.

3. Working of Tween Script



4. Steps to Use Tween Animation in Lens Studio

1. Open Lens Studio and create a new project.
2. Add a 2D or 3D object to the scene.
3. Create or add a script asset.
4. Define the starting and target values of the object.
5. Set the animation duration and easing function.
6. Start the tween animation through the supported scripting API or tween library.
7. Preview the Lens and adjust the animation as required.

5. Example of Tween Concept

Suppose a virtual ball starts at position 0 and needs to move to position 10 in 2 seconds.

```
// Starting position
var startPosition = 0;


// Target position
```

var endPosition = 10;

// Animation duration

var duration = 2;

The tween system gradually changes the ball's position from 0 to 10 over 2 seconds.

Note: This code shows the basic tween concept, not a complete Lens Studio implementation. The actual tween API and object properties depend on the scripting setup or library used.

6. Applications of Tween Script

- Object Movement: Moves virtual objects smoothly.
- Pop-up Effects: Makes objects appear or disappear smoothly.
- Scaling Effects: Enlarges or shrinks objects.
- Rotation: Rotates 3D models.
- UI Animation: Animates buttons and text.
- Interactive Lenses: Creates animations triggered by user actions.

7. Advantages

1. Makes animations smooth and attractive.
2. Reduces the need to manually update every animation frame.
3. Allows control over duration and movement style.
4. Improves the user experience.
5. Supports creative and interactive AR effects.

8. Difference Between Behaviour Script and Tween Script

Behaviour Script	Tween Script
Controls the behaviour and logic of objects.	Focuses on smooth transitions and animations.
Responds to events and conditions.	Changes values over time.
Can perform many types of actions.	Commonly animates position, scale, rotation, and other properties.

Q17. Explain Orthographic Camera in Lens Studio. (10 Marks)

1. Definition

An Orthographic Camera is a camera that displays 3D objects without perspective distortion. Objects remain the same apparent size regardless of their distance from the camera.

In Lens Studio, it can be used to create flat-looking 2D scenes, interface elements, and stylized visual effects.

2. Features of Orthographic Camera

1. No Perspective Distortion: Objects do not appear smaller when they move farther away.
2. Uniform Object Size: Objects of the same size appear equally sized at different depths.

3. Parallel Lines: Parallel lines remain parallel in the rendered view.
4. Fixed Viewing Area: The visible area is controlled by the camera's orthographic size or projection settings.
5. 2D and 3D Support: It can display 2D elements and 3D objects with a flat appearance.
6. Easy Positioning: Useful for arranging objects in a fixed, predictable layout.

3. Working of Orthographic Camera



4. Steps to Use an Orthographic Camera

1. Open Lens Studio and create a project.
2. Add the required objects to the scene.
3. Add or select a camera that supports orthographic projection.
4. Set its projection type to Orthographic, if available.
5. Adjust the camera's position and viewing size.
6. Preview the scene and check the object's appearance.
7. Test the Lens to ensure the scene displays correctly.

Note: Camera projection options and setup may vary by Lens Studio version and scene type.

5. Example

Suppose a developer creates a 2D-style game interface with several icons. With an orthographic camera, the icons maintain their apparent size even when placed at different depths, making the layout easier to control.

6. Applications of Orthographic Camera

- 2D Interfaces: Displays buttons, icons, and text.
- 2D-Style Games: Creates flat-looking game scenes.
- AR Effects: Produces stylized visual effects.
- Technical Visualization: Shows objects without perspective distortion.
- Graphic Design: Helps arrange objects in a predictable layout.
- Animation: Creates flat or cartoon-like scenes.

7. Difference Between Orthographic and Perspective Camera

Orthographic Camera	Perspective Camera

Objects do not shrink with distance.	Objects appear smaller as distance increases.
No perspective size distortion.	Creates a realistic sense of depth.
Parallel lines remain parallel.	Parallel lines may appear to converge.
Useful for flat layouts and stylized scenes.	Useful for realistic 3D scenes.
Size depends mainly on object scale and camera settings.	Apparent size depends on object size and distance.

8. Advantages

1. Provides a consistent object size in the rendered view.
2. Makes 2D-style layouts easier to design.
3. Avoids perspective distortion.
4. Helps create clean and predictable scenes.

Q18. Explain Bitmoji in Lens Studio. (10 Marks)

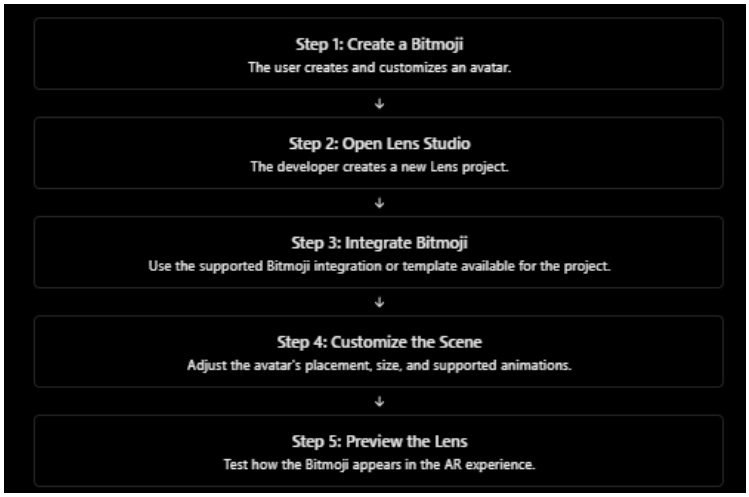
1. Definition

Bitmoji is a personalized cartoon avatar that represents a user. In Lens Studio, Bitmoji integration allows developers to use supported Bitmoji avatars in AR Lenses to create personalized and interactive experiences.

2. Features of Bitmoji in Lens Studio

1. Personalized Avatar: Represents the user through a customized cartoon character.
2. 3D Avatar: Supported integrations can display Bitmoji characters in a 3D environment.
3. Custom Appearance: The avatar reflects the user's selected Bitmoji appearance.
4. AR Integration: Allows avatars to appear in augmented reality scenes.
5. Animation Support: Supported avatars may perform animations or actions.
6. Interactive Experience: Developers can create engaging Lenses featuring personalized avatars.

3. Working of Bitmoji in Lens Studio



4. Steps to Use Bitmoji in Lens Studio

1. Open Lens Studio and create a project.

2. Check for a supported Bitmoji template or integration.
3. Add the Bitmoji resource or avatar component as required.
4. Configure the avatar's position, scale, and appearance.
5. Add a suitable background or other AR objects.
6. Preview and test the Lens.
7. Publish the Lens if all requirements are satisfied.

5. Applications of Bitmoji

- Social Media: Creates personalized Snapchat Lenses.
- Entertainment: Adds cartoon characters to AR scenes.
- Storytelling: Uses avatars in animated stories.
- Gaming: Includes personalized characters in interactive experiences.
- Communication: Represents users through cartoon avatars.
- Creative Marketing: Creates engaging branded AR experiences.

6. Example

Suppose a developer creates a Lens showing a user's Bitmoji standing beside a virtual birthday cake. The developer can design the scene with decorations and supported animations to create a personalized birthday experience.

7. Advantages of Bitmoji

1. Makes AR experiences more personal.
2. Provides an attractive cartoon appearance.
3. Improves user engagement.
4. Supports creative and entertaining Lenses.
5. Helps create memorable social media content.

Q19. Explain Generative AI and Its Types. (10 Marks)

1. Definition

Generative Artificial Intelligence (Generative AI) is a type of AI that creates new content by learning patterns from existing data. It can generate text, images, audio, video, and computer code.

Examples: ChatGPT generates text, DALL·E generates images, and AI music tools generate audio.

2. Types of Generative AI

1. Text Generation

- Generates human-like text from user instructions.
- Used for answering questions, writing articles, and summarizing information.
- Example: ChatGPT.

2. Image Generation

- Creates new images from text prompts or reference images.
- Used for graphic design, digital art, and advertising.

- Example: DALL·E.

3. Audio and Music Generation

- Generates speech, sound effects, and music.
- Used in voice assistants, music production, and dubbing.
- Example: AI music-generation tools.

4. Video Generation

- Creates videos from text prompts, images, or other inputs.
- Used in filmmaking, animation, and marketing.
- Example: AI text-to-video tools.

5. Code Generation

- Generates computer programs or suggests code based on instructions.
- Helps developers write, explain, and debug programs.
- Example: GitHub Copilot.

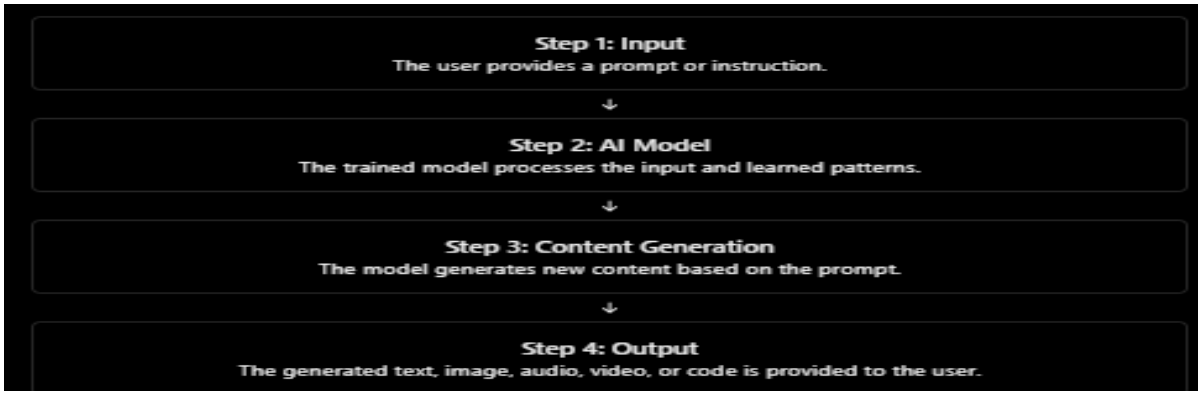
6. 3D Model Generation

- Creates three-dimensional objects and models from text or images.
- Used in AR/VR, gaming, animation, and product design.
- Example: AI tools that generate 3D objects for virtual environments.

3. Main Technologies Used in Generative AI

1. Generative Adversarial Networks (GANs): Use two neural networks—a generator and a discriminator—to create realistic data.
2. Variational Autoencoders (VAEs): Learn compressed representations of data and generate new samples from them.
3. Transformers: Learn relationships in sequences and are widely used for text and code generation.
4. Diffusion Models: Generate images and other content by gradually removing noise from random data.

4. Working of Generative AI



5. Applications of Generative AI

- Education: Creates notes, explanations, and practice questions.

- Healthcare: Supports medical research and synthetic data generation.
- Software Development: Generates and explains code.
- Entertainment: Creates images, music, animation, and videos.
- Business: Generates advertisements, reports, and product descriptions.
- AR/VR: Helps create virtual objects, environments, and interactive content.

6. Advantages and Limitations

Advantages

1. Saves time and increases productivity.
2. Supports creativity and content creation.
3. Automates repetitive tasks.
4. Helps generate ideas and prototypes quickly.

Limitations

1. May produce incorrect or misleading information.
2. Can reflect bias present in training data.
3. May raise privacy and copyright concerns.
4. Generated content may require human review.

Q1. Explain AR and VR.

Q2. Explain Different Types of AR and VR.

Q3. Explain Applications and Use Cases of AR and VR.

Q4. Explain Lens Studio and Steps to Publish a Lens.

Q5. Explain Lens Size, Text Size and App Icon in Lens Studio.

Q6. Explain How to Add Text in Lens Studio.

Q7. Explain How to Add Audio and Video in Lens Studio.

Q8. Explain World Mesh in Lens Studio.

Q9. (Question not provided in this conversation.)

Q10. Explain Hand Tracking in Lens Studio.

Q11. Explain Hand Joints in Lens Studio.

Q12. Explain the Difference Between 2D and 3D Hand Tracking.

Q13. Explain Body Tracking in Lens Studio.

Q14. Explain Body Mesh in Lens Studio.

Q15. Explain Behaviour Script in Lens Studio.

Q16. Explain Tween Script in Lens Studio.

Q17. Explain Orthographic Camera in Lens Studio.

Q18. Explain Bitmoji in Lens Studio.

Q19. Explain Generative AI and Its Types.